WORKED SOLUTION

STEP II 2001, Question 4

Pure • Multiple Angle Equations

STEP II 2001, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q4

The key idea

Every term of f has a factor $\sin x$: $\sin 2x = 2\sin x \cos x$, and the hint does the same for $\sin 3x$. Pulling out $\sin x$ leaves a quadratic in $\cos x$, and the condition on P, Q, R is exactly "this quadratic has negative discriminant". The second part is the same function with $P = 1$, $Q = 1$, $R = -1$ and x replaced by $2nx$. Now the quadratic does have roots, and the work is in finding which root sits closest to the top of the allowed range.

Solution

The first result

Write $c = \cos x$. Using $\sin 2x = 2\sin x \cos x$ and the given identity for $\sin 3x$,

$$\begin{aligned} \mathrm{f}(x) &= P\sin x + 2Q\sin x\cos x + R\sin x\,(4\cos^2 x - 1) \\ &= \sin x\,(4Rc^2 + 2Qc + (P - R)). \end{aligned}$$

So $\mathrm{f}(x) = 0$ exactly when $\sin x = 0$ or $4Rc^2 + 2Qc + (P - R) = 0$.

First, $R \neq 0$: if $R = 0$ the condition would read $Q^2 < 0$, which is impossible. So the bracket is a genuine quadratic in c, and its discriminant is

$$(2Q)^2 - 4 \cdot 4R\,(P - R) = 4(Q^2 - 4R(P - R)) < 0$$

by hypothesis. A real quadratic with negative discriminant has no real roots, so the bracket is never zero for any real c, and in particular not for $c = \cos x$. Hence $\mathrm{f}(x) = 0$ forces $\sin x = 0$, that is, $x = m\pi$ with m an integer. Conversely, every $x = m\pi$ is a zero of f, so these are precisely the zeros, as required.

The largest root of g

Put $y = 2nx$. Then $\mathrm{g}(x) = \sin y + \sin 2y - \sin 3y$, which is $\mathrm{f}(y)$ with $P = 1$, $Q = 1$, $R = -1$. Here $Q^2 = 1$ is not less than $4R(P - R) = -8$, so the first result does not

apply, and the quadratic has real roots. With $c = \cos y$, the factorisation above gives

$$g(x) = \sin y\,(-4c^2 + 2c + 2) = -2\sin y\,(2c+1)(c-1).$$

So $g(x) = 0$ exactly when

$$\sin y = 0, \qquad \cos y = 1, \qquad \text{or} \qquad \cos y = -\tfrac{1}{2}.$$

The case $\cos y = 1$ is already contained in $\sin y = 0$. So the roots in y are

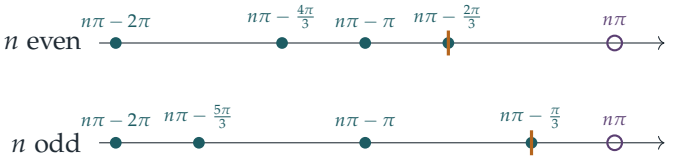
$$y = k\pi \quad \text{or} \quad y = 2k\pi \pm \tfrac{2\pi}{3} \qquad (k \in \mathbb{Z}).$$

Within each period $[2k\pi, 2k\pi + 2\pi)$ the roots sit at $2k\pi$ plus 0, $\frac{2\pi}{3}$, π, $\frac{4\pi}{3}$.

The range $0 < x < \frac{1}{2}\pi$ becomes $0 < y < n\pi$, and the right-hand end $y = n\pi$ is itself a root but is excluded. So we want the largest root strictly below $n\pi$, and only the position of $n\pi$ within its period matters.

Ask yourself Why does the parity of n matter?

The pattern of roots repeats every 2π, not every π. If n is even, $n\pi$ is a multiple of 2π; if n is odd, $n\pi$ is an odd multiple of π. In the two cases the roots just below $n\pi$ are spaced differently.



(Roots of g in terms of y over the last 2π below $n\pi$, drawn to scale. The open circle marks the excluded end and the amber tick marks the largest root.)

n even. Then $n\pi = 2k\pi$ with $k = \frac{n}{2} \geq 1$, and the roots in the last period below it are, measured from $n\pi$,

$$n\pi - 2\pi, \quad n\pi - \tfrac{4\pi}{3}, \quad n\pi - \pi, \quad n\pi - \tfrac{2\pi}{3}.$$

The last of these is $2(k-1)\pi + \frac{4\pi}{3}$, a root because its cosine is $-\frac{1}{2}$. There is no root in $(n\pi - \frac{2\pi}{3}, n\pi)$: no multiple of π lies there, and $\cos y = -\frac{1}{2}$ would need $y \equiv \frac{2\pi}{3}$ or $\frac{4\pi}{3}$ $(\text{mod } 2\pi)$, whereas here y lies strictly between $\frac{4\pi}{3}$ and 2π mod 2π. Also $n\pi - \frac{2\pi}{3} > 0$ since $n \geq 2$. So the largest root has $y = n\pi - \frac{2\pi}{3}$, and

$$x = \frac{y}{2n} = \frac{(3n-2)\pi}{6n} \qquad (n \text{ even}).$$

n odd. Then $n\pi = 2k\pi + \pi$ with $k = \frac{n-1}{2} \geq 0$. Measured within the period starting at $2k\pi = n\pi - \pi$, the roots below $n\pi$ are at offsets 0 and $\frac{2\pi}{3}$, so the candidates are $n\pi - \pi$ and $n\pi - \frac{\pi}{3}$. The interval $(n\pi - \frac{\pi}{3}, n\pi)$ corresponds to y strictly between $\frac{2\pi}{3}$ and π mod 2π, which contains no root. And $n\pi - \frac{\pi}{3} \geq \frac{2\pi}{3} > 0$. So the largest root has $y = n\pi - \frac{\pi}{3}$, and

$$x = \frac{(3n-1)\pi}{6n} \qquad (n \text{ odd}).$$

Checks

For $n = 1$, $\mathrm{g}(x) = \sin 2x + \sin 4x - \sin 6x$ on $(0, \frac{1}{2}\pi)$: the formula gives $x = \frac{\pi}{3}$, and indeed $\sin\frac{2\pi}{3} + \sin\frac{4\pi}{3} - \sin 2\pi = 0$. For $n = 2$ the formula also gives $\frac{\pi}{3}$ ($y = \frac{4\pi}{3}$). As $n \to \infty$ both formulae tend to $\frac{1}{2}\pi$, as they should, since the roots crowd together. We scanned g numerically on a fine grid for $n = 1, \ldots, 12$. Each predicted value (for example $x = \frac{4\pi}{9} \approx 1.3963$ for $n = 3$) is a root, and g keeps one sign between it and $\frac{1}{2}\pi$. For the first part, 2000 random triples satisfying $Q^2 < 4R(P - R)$ all gave a bracket with no zero on $[-1, 1]$.

Common slips

- Giving $x = \frac{1}{2}\pi$ itself: it is a root of g for every n, but it is outside the open interval.
- Forgetting the roots from $\sin y = 0$ and keeping only $\cos y = -\frac{1}{2}$. For even n, $y = n\pi - \pi$ is a root, though not the largest.
- Writing $\cos y = -\frac{1}{2} \Rightarrow y = \frac{2\pi}{3} + 2k\pi$ only, and missing the $\frac{4\pi}{3}$ family. That gives the wrong answer for even n.

Beyond STEP

The factorisation works because $\sin mx / \sin x$ is always a polynomial in $\cos x$: these are the Chebyshev polynomials of the second kind, $U_{m-1}(\cos x)$. So any finite sine series $\sum a_m \sin mx$ is $\sin x$ times a polynomial in $\cos x$, and counting its zeros becomes counting polynomial roots in $[-1, 1]$. This idea underlies much of approximation theory and Fourier analysis. Needham's *Visual Complex Analysis* gives a lovely view of multiple-angle formulae through $\mathrm{e}^{\mathrm{i}x}$.

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